

Wrist Tendinopathies

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Tendinopathies are extremely common in athletes. Any sport that requires a repetitive motion, such as a tennis serve or volley, a basketball free throw, or “turning the wrists over,” as in the completion of a golf or baseball swing, puts an athlete at risk for tendon inflammation, instability, degeneration, or even rupture.^{1,2} Even recreational rock climbing, which is well known for its association with flexor pulley rupture, has been implicated in tendinopathy of the wrist.^{3,4} Swimming, bowling, gymnastics, weightlifting, cycling, and skiing are among the many other sports associated with repetitive use tendinopathy.⁵

Montalvan et al.⁶ categorized extensor carpi ulnaris (ECU) tendinopathy in 28 elite tennis players. The authors divided ECU tendon injuries into three categories: tendinopathy, instability, or rupture. Allende and Le Viet⁷ reported on 28 patients who underwent surgical treatment for ECU tendinopathy between 1990 and 2002. Seventeen of the 28 patients reported onset of symptoms after sports activity. Of the 28 patients, 15 had tenosynovitis or tendinosis, 5 had dislocation of the tendon, 4 had subluxation, and 4 had a tendon rupture.⁷ Twenty-two of the 28 patients were able to return to their previous level of activity at a mean of 23 months after surgery.

The first dorsal compartment, which contains the abductor pollicis longus (APL) and extensor pollicis brevis (EPB) tendons, is also subject to wrist tendinopathy, specifically de Quervain tenosynovitis, especially in persons who participate in sports involving a racquet.⁸

Soejima et al.⁹ reported a case of flexor carpi radialis (FCR) tendinopathy in a professional baseball player that was associated with a malunited trapezial ridge fracture.⁹ The patient was successfully treated with excision of the trapezial ridge. Buterbaugh and colleagues,¹⁰ in evaluating ulnar-sided wrist pain in seven hockey or baseball professional athletes, reported acute calcific flexor carpi ulnaris (FCU) tendinitis as one of the contributing sources. Some sports that require repetitive wrist extension and radial deviation, such as rowing and powder skiing, have been found to be associated with intersection syndrome.^{5,8,11}

Even the extensor retinaculum, the soft tissue layer that provides a sheath-like investing roof for the extensor tendons over the dorsal aspect of the distal radius and carpus, has been implicated in athletes as a source of pain. Thickening and diffuse proliferation of this tissue may result from hyperextension activities required in certain athletic pursuits. This particular scenario has been termed *extensor retinaculum impingement*.¹²

For all of the tendinopathies described in sports, early diagnosis and nonoperative treatment with activity modification, use of a splint or brace, and use of nonsteroidal antiinflammatory drugs (NSAIDs) has been the cornerstone of management. This chapter presents a discussion of the history and physical examination and the common tendinopathies associated with sports, as well as treatment options, with return to sports as the ultimate goal.

HISTORY

The history is a critical component in making a correct diagnosis. Tendons are the structures that connect muscle bellies to bones and allow for confluent joint and extremity motion. Therefore any repetitive wrist or hand motions that elicit pain are an important focus in a person with a suspected tendinopathy. If active wrist flexion causes pain, then the flexor tendons may be inflamed. Similarly, if active extension is painful, then extensor tendons may be implicated in the pathology. Conversely, passive stretching of an irritated tendon also causes pain and can be a clue as to which tendon, or tendons, are involved.

The location of the pain is very important. Often when tendinopathy is present, an athlete runs his or her hand longitudinally along the tendon(s) involved, offering a key diagnostic clue. A fracture is often painful at a specific place, whereas tendons tend to “hurt” along the longitudinal course of the tendon. At this point, it is important to note that “pain” is a historic, or subjective, complaint, whereas “tenderness” is a component of the physical examination in which the examiner elicits pain through palpation or pressure.

Next, the timing of the injury is important. Is the pathology the result of repetitive activity, or did it occur after one specific incident? For example, a tendon rupture typically occurs suddenly, after a single event, whereas tendinosis tends to be the result of repetitive movements of the hand or wrist, such as is seen in ping pong, tennis, or sports that involve throwing. Often with tendinosis, the pain recurs with the specific exacerbating motion, such as flexion, extension, radial or ulnar deviation, and, in some cases, pronation and supination of the wrist. All of these questions should be included in the history to help elucidate the correct diagnosis. Any exacerbating or alleviating events should also be ascertained. It is important to know if placing the wrist in a certain position or if performing certain motions or maneuvers worsen (or lessen) the pain; these facts

Abstract

Tendinopathies are common in patients with vocations or hobbies that involve repetitive motions. Given the complex anatomy of the hand and wrist, a thorough history and physical exam is critical to identify the underlying pathology. History should elicit activities with repetitive wrist motions, as well as prior treatments attempted. Physical examination should assess for asymmetric swelling, and include special tests such as the Finkelstein and extensor carpi ulnaris (ECU) synergy tests. Imaging at the minimum, should consist of plain radiographs with additional information offered by magnetic resonance imaging (MRI) and nuclear medicine testing.

The above history and physical exam often leads to the diagnosis of several common tendinopathies seen in clinical practice. These include acute and chronic tendinopathies affecting the extensor carpi ulnaris, first dorsal compartment, the proximal intersection of the first and second dorsal compartments, flexor carpi radialis, and flexor carpi ulnaris.

After identification of a tendinopathy, oftentimes the first-line treatment is nonoperative management. This consists of activity modification, ice, oral antiinflammatories, and soft tissue wrist via splinting. When these measures are unsuccessful, surgical release or rerouting may be considered. Return to play and vocation is largely guided by pain free in nonoperative management.

Keywords

hand
wrist
tendinopathy
acute tendon injury
intersection syndrome
de Quervain tenosynovitis

also may provide a clue to the diagnosis. Any prior treatments, no matter how menial, are important historic data. If therapy has been initiated, which is often the case in high-level athletics, it is important to distinguish between exercises or modalities that have proven beneficial or detrimental to recovery. For example, stretching programs may exacerbate the symptoms of tendinopathy when they are implemented before tendon healing, which helps lead the interviewer to the correct diagnosis.

PHYSICAL EXAMINATION

Direct evaluation of the wrist should always begin with inspection. Any visible swelling or asymmetry when compared with the other wrist is a very important clue in diagnosing wrist tendon pathology. In the case of de Quervain tenosynovitis, which involves the tendons of the first dorsal compartment, it is quite common for swelling to occur over the radial aspect of the wrist at the level of the radial styloid. When the wrists are placed side by side, the asymmetric swelling often becomes evident.

The hallmark of physical examination in persons with acute calcific tendinitis is severe tenderness to palpation along the course of the tendon. Whereas tenderness in a fracture occurs at the point of fracture, tenderness associated with tendinosis is often more diffusely felt along the course of the involved tendon. The exception may be insertional tendinopathy, in which case the maximal tenderness is elicited at the insertion point of the involved tendon onto bone. The pain is exacerbated by passive stretch of the involved tendon, further aiding diagnosis. Arthritis pain, conversely, tends to increase with compression across a joint, whereas passive stretching may actually alleviate the pain associated with arthritis or joint pathology.

Certain tests have been determined to be associated with specific tendon pathology. The Finkelstein test, in which the examiner grasps the patient's thumb and ulnarly deviates the wrist to passively stretch the APL and EPB tendons—that is, the tendons of the first dorsal compartment—aids in diagnosing de Quervain tenosynovitis. This maneuver will reproduce pain in most patients with this condition.

Other palpation techniques include the detection of tendon snapping or popping during wrist motion. For example, the ECU tendon may snap over the ulnar styloid when the extended wrist is brought actively from a pronated to a fully supinated position. The patient often feels the tendon snap with the supination maneuver, and the examiner's index finger, when lightly placed over the ulnar styloid, may confirm the abnormal tendon subluxation or frank dislocation.

Range of motion (ROM) may be diminished in tendinopathies such as wrist extension in the case of flexor tendon irritation. To comprehensively assess motion, the patient should be asked to flex and extend, radially and ulnarly deviate, and pronate and supinate the wrist in sequence. Repeating these motions against resistance may reveal subtle tendon pathology. Because variations of normal exist, comparing the ROM with the contralateral wrist is necessary to detect loss of motion in any direction.

In addition to the tendinitis, deciding whether bone, joint, ligament, nerve, or even a vascular structure is concurrently injured depends on the examiner's ability to synthesize appropriate history

clues with key physical examination findings. This approach will help the practitioner avoid compartmentalizing every athletic wrist complaint as a tendon-related pathology and steer him or her toward appropriate adjunctive imaging studies.

IMAGING

Tendinopathy of the wrist is not a bony pathology, but plain radiographs and advanced imaging, when indicated, can help determine and finely hone diagnostic evaluations of the wrist.¹³ Imaging can either eliminate or support the diagnosis of fracture, carpal instability, and Kienböck disease, for example, which are in the differential diagnosis of the athlete with wrist pain (Fig. 71.1).

At the minimum, imaging should include two orthogonal views; a posteroanterior (PA) view and a lateral view of the wrist should be obtained. A 45-degree semipronated oblique view and a PA view in ulnar deviation to maximally delineate the scaphoid are also valuable in examining for scaphoid pathology.¹⁴ A clenched-fist anteroposterior view is a form of stress view that may suggest a scapholunate ligament injury and potential carpal instability. Whenever doubt exists, a radiograph of the contralateral wrist is warranted, because widening of the scapholunate interval, for example, may be a bilateral finding and not indicative of a specific injury. In addition, in-office imaging with ultrasound may be of benefit. Ultrasound has the advantage over radiographs in its ability to dynamically assess for tendon pathology including ECU instability, but is highly operator dependent.^{15,16}

Additional studies should be directed by history and physical examination findings and should be used to confirm or deny the presence of a suspected injury rather than as a “shotgun” element of the workup. A MRI study, with or without intra-articular administration of gadolinium, may be performed to assess the presence of intrinsic ligamentous or triangular fibrocartilage complex (TFCC) injury.¹⁷ Gadolinium injected into the radiocarpal joint may be seen to leak into either the midcarpal joint or into the distal radioulnar joint if a significant ligament injury is present. This magnetic resonance arthrogram (MRA) is an extremely useful tool when correlated with history, physical examination, and plain radiographic findings that suggest a specific injury. However as with any imaging modality, MRA is not 100% sensitive, and may be negative in the setting of true pathology.¹⁸ Thus imaging must be taken into account with physical exam when determining clinical significance.

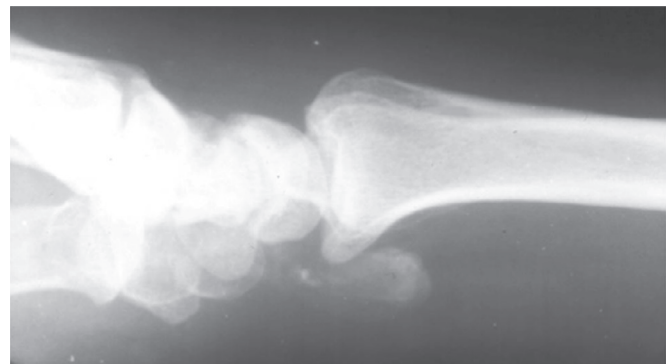


Fig. 71.1 Plain radiograph demonstrating acute calcific tendinitis.

Nuclear medicine studies such as bone scintigraphy (a bone scan) can be helpful as a sensitive indicator of acute injury, such as occult scaphoid fracture, but may lack the specificity to define an exact injury.¹⁹ Bone scans should be ordered no less than 24 to 48 hours after injury to eliminate the possibility of false-negative results. Practically speaking, given the widespread availability of MRI, bone scans have fallen out of favor as a preferred imaging modality for wrist pathology.

COMMON WRIST TENDINOPATHIES

De Quervain Tenosynovitis

Tendinopathy of the first dorsal compartment involves the APL and EPB tendons. It is more common in women than in men, as has been associated with pregnancy and the postpartum period, but has also been noted with certain sports, especially rowing and racquet sports.^{20–24} De Quervain tenosynovitis has also been described in volleyball players and should be suspected in any athlete who uses repetitive forceful wrist motions.²⁵ The area deep to the extensor retinaculum, which overlies the radial styloid, may be swollen and markedly tender. When performing Finkelstein's test, the examiner holds the patient's thumb inside a closed fist and the examiner then passively, ulnarly deviates the wrist. This maneuver reproduces pain in persons with de Quervain tenosynovitis. In the Eichhoff test, the patient is asked to clasp his or her thumb inside the flexed fingers of the hand and then ulnarly deviate the wrist (Fig. 71.2). Specific tenderness is elicited when digital pressure is placed over the radial styloid area.

Imaging can be helpful in the diagnosis of de Quervain tenosynovitis. Plain radiographs can reveal the presence of a bony spur in the floor of the first dorsal compartment. MRI can demonstrate both intrinsic tendon changes and extrinsic inflammation (Fig. 71.3). Ultrasound may demonstrate thickening involving the tendons and their sheaths as well as any instability.¹⁶

Treatment Options

The treatment of de Quervain tenosynovitis begins with immobilization in a thumb spica splint or cast. NSAIDs are also useful. Injection of a corticosteroid into the first dorsal compartment is another option although patients should be cautioned over



Fig. 71.2 Eichhoff test. With the thumb clasped into the palm, ulnar deviation of the wrist produces pain, or a positive test.

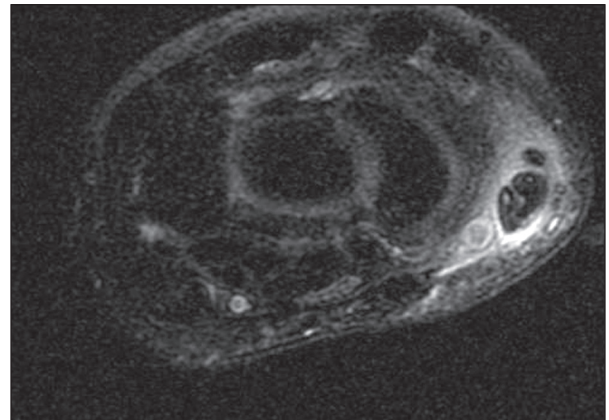


Fig. 71.3 A magnetic resonance imaging T2-weighted scan shows increased signal intensity around the tendons of the first dorsal compartment, indicating de Quervain tendinitis.

the risk of skin hypopigmentation and atrophy if placed too superficial. Avci and colleagues²¹ compared the efficacy of cortisone injection with that of splinting in pregnant or lactating women and found that cortisone injection was superior in alleviating symptoms. Lastly, when these treatments are not effective, surgical release of the first dorsal compartment may be indicated. During this procedure, care must be taken to identify both the APL and EPB based on tendon actions.

Authors' Preferred Technique

De Quervain Tenosynovitis

We prefer a stepwise treatment plan for persons with de Quervain tenosynovitis. The diagnosis is largely clinched by history and physical examination findings, as previously noted. Radiographs may help distinguish de Quervain tenosynovitis from problems related to the thumb carpometacarpal, scaphotrapezotrapezoidal, or radiocarpal joints. Our initial treatment is oral administration of an NSAID, coupled with use of a thumb spica splint or cast and activity modification. We provide a corticosteroid injection if a patient continues to have symptoms after 4–6 weeks of wearing a splint. We use 1 mL of dexamethasone (10 mg/mL) mixed with 1 mL of 1% lidocaine without epinephrine. Patients who have persistent pain despite 3 months of nonoperative management are offered elective release of the first dorsal compartment (Fig. 71.4). We prefer making a 2-cm transverse incision centered approximately 1 cm proximal to the radial styloid. Care is taken to identify and protect branches of either the superficial branch of the radial nerve

or the lateral antebrachial cutaneous nerve, which are inevitably encountered during the subcutaneous dissection. The investing extensor retinaculum overlying the first compartment is incised from distal to proximal under direct visualization, cheating dorsally (i.e., staying more dorsal to the midaxial line with the retinacular incision) thereby preventing volar subluxation of the tendons with wrist flexion. We ensure that the tendons are fully released by using a small retractor to pull on the tendons, ascertaining that both the APL and EPB tendons have been released independently. Based on passive extension of the thumb metacarpal and proximal phalanx when the appropriate tendon is pulled, full release of both tendons is thus verified. We believe this step is important because the APL may have multiple tendon slips, and a vertical septum may separate the smaller EPB tendon into a separate "chamber" within the first dorsal compartment. These anatomic aberrancies have been historically cited as predisposing persons to this condition.

Continued

Authors' Preferred Technique

De Quervain Tenosynovitis—cont'd

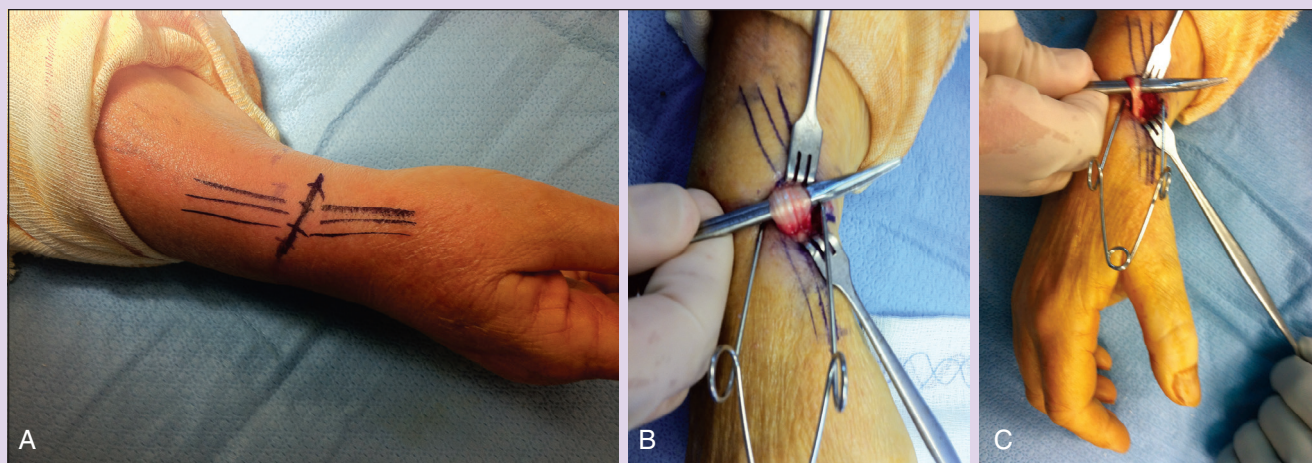


Fig. 71.4 (A) Planned incision for surgical release of the first dorsal compartment for de Quervain tenosynovitis. (B) The abductor pollicis longus (APL) tendon is visible in the incision once the overlying retinaculum is incised. Notice that three separate tendon slips are visible in this case. (C) The extensor pollicis brevis tendon is smaller than the APL but must be released as well. Pulling on the tendon should cause extension of the proximal phalanx of the thumb.

Postoperative Management and Return to Sports

A plaster thumb spica splint is placed immediately after surgical treatment for de Quervain tenosynovitis, and the patient is seen 10 to 14 days after surgery for suture removal and placement of a removable thumb spica splint. Athletes who require surgical release for refractory de Quervain tenosynovitis typically require 6 weeks of postoperative splinting to allow symptoms to resolve. Return to sports is allowed once ROM of the thumb and strength return. For athletes who do not require surgery, return to sports is permissible with use of a functional brace or cast when practical. Use of a functional brace or cast may be virtually impossible in sports such as basketball or gymnastics that require freedom at the wrist. Otherwise, athletes are allowed to return to sports when symptoms resolve and return of motion and strength are demonstrated.

Intersection Syndrome

Intersection syndrome has been associated with sports that involve repetitive extension and radial deviation of the wrist, such as skiing, weightlifting, rowing, and racquet sports.^{5,8,11,22} Radial-sided dorsal wrist pain is the hallmark symptom, but it is typically perceived as more proximal than the pain associated with de Quervain tenosynovitis (Fig. 71.5).²⁶ Athletes may refer to intersection syndrome as “crossover tendinitis” because it occurs in the region of the distal forearm where the tendons of the first dorsal compartment intersect, or cross over, the tendons of the second dorsal compartment (extensor carpi radialis longus and brevis).⁵ The inflammatory component is located within the second dorsal compartment. Clinical findings include localized tenderness and, in more severe cases, crepitus with wrist extension at this intersection, approximately 4 cm proximal to the wrist joint. There may be a “wet leather sign” with wrist extension versus resistance, where a sound often described as the one made

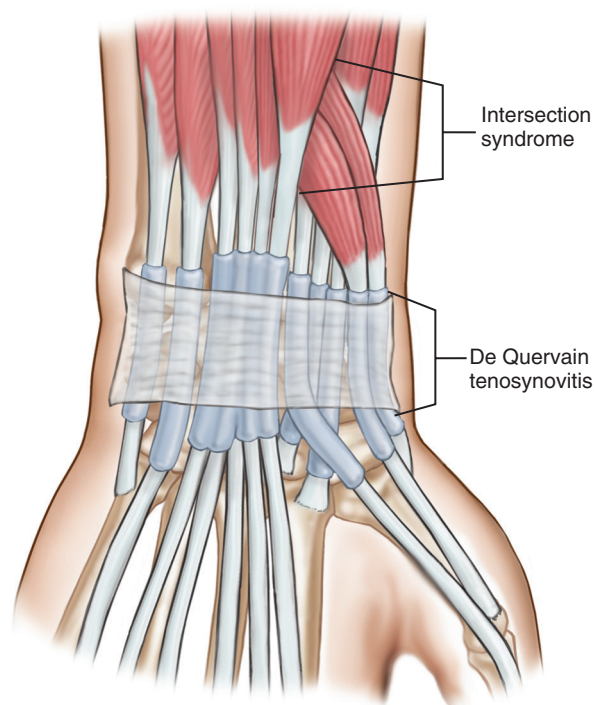


Fig. 71.5 The specific area of tenderness elicited in persons with intersection syndrome is typically more proximal than that of de Quervain tenosynovitis, as shown in this anatomic drawing.

when rubbing wet leather is heard. MRI is a useful adjunct in making the diagnosis, with increased peritendinous signal evident on T2-weighted images at the offending location.²⁷

Treatment Options

Treatment is directed at resting the involved tendons in a thumb spica cast or brace, with oral administration of NSAIDs and

cessation of the exacerbating activity. Grundberg and Reagan²⁸ demonstrated that the pathologic anatomy is a stenosing tenosynovitis of the investing sheath of the extensor carpi radialis longus and extensor carpi radialis brevis tendons. Refractory cases may require release of the stenotic sheath of the second dorsal compartment and débridement of associated tendinosis and hypertrophic tenosynovium.²⁸



Authors' Preferred Technique

Intersection Syndrome

Rest, ice, and a forearm-based thumb spica splint are the cornerstones of management. NSAIDs are also prescribed. Corticosteroid injections are infrequently necessary. Use of nonoperative management without the need for surgery has been the experience of most wrist specialists treating this condition. Though we have never had to surgically treat intersection syndrome, we agree with *Green's Operative Hand Surgery*²⁰ adaptation of the Grundberg and Reagan approach to surgical treatment of intersection syndrome. In this method, a longitudinal incision is made from the area of swelling distally to the radial wrist extensors. The deep fascia overlying the second dorsal compartment is then released and the tendons débrided of any tenosynovitis.

Return to Sports

Athletes are allowed to return to sports once the symptoms have dissipated, or alternatively while still wearing a thumb spica cast or splint if practical for the sport in question.

Extensor Carpi Ulnaris Tendinopathy

Situated in the sixth dorsal compartment at the ulnar aspect of the wrist, the ECU tendon is commonly compromised by sporting activities. Golfers, tennis players, and other participants in racquet sports are at particular risk for injuries to the ECU.^{6,7,29,30} ECU tendon problems may manifest as inflammatory tendinitis, degenerative tendinosis, instability (subluxation or dislocation), or rupture.⁷ One specific form of tendinopathy, acute calcific tendinitis, presents with intense pain in the area of the involved tendon, and the etiology that differentiates it from degenerative tendinosis is poorly understood.²⁰

Anatomically the ECU is adjacent to the ulnar styloid, and dorsal-to-palmar dislocation or subluxation of the tendon can lead to a painful snapping sensation when the wrist undergoes simultaneous extension and rotation.^{1,21} Because the floor of the ECU sheath is inherently considered a part of the TFCC, tendinopathy can and does coexist with TFCC injury in some cases.²⁹ Given the aforementioned complex, and potentially multifaceted, etiology of ulnar-sided wrist pain, Kakar et al. recently described a “four-leaf clover” diagnostic algorithm to guide treatment. The algorithm consists of four questions including whether the ECU is unstable, if there is a concomitant TFCC injury, and if any chondral defect or bony deformity exists. This algorithm allows surgeons to ensure all components of ulnar-sided wrist pain are addressed. In the case of multiple positive answers to the above questions, treatment consists of addressing each of these injuries.³¹

Classification

Differentiating between three types of tendon pathology for the ECU is useful in directing treatment. As previously stated,

Montalvan et al.⁶ classified ECU tendon pathology into three categories: acute instability, intrinsic tendinopathy, and rupture. The authors indicate that treatment differs for each of the tendon pathology subtypes, making the classification relevant and important. For any athlete who presents with acute onset of ulnar-sided wrist pain, ECU tendinopathy is high on the list of differential diagnoses. Because it may exist in the presence of other pathology, such as TFCC injury, the exact etiology may be elucidated using the aforementioned “four-leaf clover” diagnostic algorithm.

Occasionally, swelling over the ECU tendon sheath is present, and active extension with ulnar deviation of the wrist elicits pain. The practitioner should inquire about a history of snapping, especially when it is associated with twisting movements of the wrist. Any sudden loss of motion, especially extension combined with ulnar deviation, may provide evidence that rupture is possible.

Special Tests

Tenderness in the area of the ulnar head with circumduction of the wrist, and pronation and supination with the wrist extended, suggests a component of ECU tendinopathy. Tenderness along the ECU tendon should be expected. Any instability of the tendon, as mentioned in the physical examination section, should also be tested. In order to do this, the examiner may feel the ECU become unstable during resisted flexion, ulnar deviation and supination of the wrist. A painful snap indicates ECU instability. Additionally pain with resisted adduction of the thumb and long finger with the forearm in full supination and the wrist in neutral extension, otherwise known as the ECU Synergy Test, is a highly sensitive and specific method of diagnosing ECU tendonitis, and helps differentiate other etiologies of pain including TFCC tears.³² Lastly, an inability to extend and ulnarly deviate the wrist may be an indication of rupture.

Treatment Options

For cases of acute tendonitis or degenerative tendinosis without instability, immobilization in a short-arm cast, splint, or brace is warranted, along with the administration of oral NSAIDs. Selective use of corticosteroid injection for symptoms unresponsive to splinting and oral medications can be attempted, but patients need to be warned about the theoretical risk of tendon rupture.⁷

Early immobilization for instability involves the use of a long-arm splint or cast with the wrist in pronation, extension, and slight radial deviation, because the supinated wrist, when brought from extension into ulnar deviation and flexion, may elicit the painful snap.²⁰ If splinting fails to alleviate subjective and objective instability, surgical reconstruction of the ECU tendon sheath to reroute and stabilize the tendon radial and dorsal to the ulnar head is justified.³³ For tendon rupture, primary repair may be needed to rebalance wrist extension and maximize function in the competitive athlete.

Criteria for Return to Sports

After nonoperative management, the timing for return to sport is based on symptom resolution, which may vary from 1 or 2 weeks to 3 months in some cases. For surgical release, a simple Z cut is made in the ECU tendon sheath to release investing

compartment. This is followed by loose approximation of the sheath to prevent instability. Postoperatively the patient is splinted for 2 weeks followed by early therapy directed at restoring motion and strength. When the patient demonstrates full painless motion and adequate strength similar to that of the contralateral side, return to sport is allowed, typically 6 weeks after surgery.

In the case of ECU tendon sheath reconstruction, a long-arm splint is used for 6 weeks after surgery, followed by an additional 6 weeks of therapy (ROM, strengthening, and proprioception training) before the athlete is allowed to return to his or her sport. Resolution of pain and instability, coupled with near-normal strength (>80% compared to the contralateral side) and motion, are the benchmarks for permitting a safe return to sports. Recovery time varies per individual. The pain-free swing of a baseball bat is likely to take far longer to recover compared to that of a ping-pong-paddle swing, and as such recovery time can be as short as 6 weeks, and up to 3 months. If it is practical for the given sport, intermittent use of a protective short-arm wrist brace is encouraged until confidence in the strength of the wrist is restored.

Flexor Carpi Radialis Tendinopathy

Tendinopathy of the FCR is distinguished by its involvement at the volar radial aspect of the wrist, with exacerbation by flexion and radial deviation of the wrist against resistance.³⁴ The FCR tendon is closely associated with the trapezium, scaphoid, and trapezoid; it courses in a fibro-osseous tunnel through the carpus and is especially vulnerable to degenerative changes as it courses

adjacent to the ridge of the trapezium on its way to insertion on the volar aspect of the bases of the index and sometimes middle metacarpals.^{35,36} Arthritic conditions of the scaphoid, trapezium, or trapezoid can lead to FCR tendinosis and even rupture.³⁷ Soejima et al.⁹ described FCR tendinopathy that was associated with a malunited trapezial ridge fracture in a professional baseball player. Resolution of the condition required excision of the malunited trapezial ridge. Most commonly, repetitive wrist motion, especially flexion with radial deviation, can lead to a tendinopathy of the FCR.³⁴ Treatment varies according to the exact etiology, but initial splint treatment with use of oral NSAIDs is usually indicated. Conservative treatment with activity modification, splints, and nonsteroidal agents is highly successful in the treatment of primary FCR tendinopathy. Corticosteroid injections are usually unnecessary in this patient population. The use of a thumb spica splint for FCR tendinopathy is not necessary. In the unusual case that it is unresponsive to conservative care, carefully planned surgical release of the FCR tunnel, along with any other offending pathology, such as trapezial ridge malunion, can be extremely beneficial.³⁴

Flexor Carpi Ulnaris Tendinopathy

In an athlete presenting with ulnar-sided wrist pain, tendinopathy of the FCU should be considered in the differential diagnosis. The clinical presentation of volar ulnar wrist pain, exacerbated with active flexion and ulnar deviation, is classic. The possibility of other etiologies, such as TFCC tears, fractures of the hook of the hamate or pisiform, lunotriquetral ligament injury, and even

Authors' Preferred Technique

Extensor Carpi Ulnaris Tendinopathy

In all cases of ECU tendinopathy without instability, we prefer to immobilize the tendon in a short-arm brace or cast for a period of 4 weeks, along with prescription of an oral NSAID. This approach is typically effective for cases of early degenerative tendinosis or acute calcific tendonitis. If splinting and medication are ineffective, we offer a corticosteroid injection, followed by an additional period of protective splinting. We prefer to use 1 mL of 10 mg/mL dexamethasone mixed with 1 mL of 1% lidocaine without epinephrine. Patients are counseled about the possibility of tendon rupture after a steroid injection and are free to decline this treatment. Surgery is warranted only in recalcitrant cases unless persistent instability is present. Surgical release of stenosing tenosynovitis typically brings a good degree of relief.⁷ We use an incision that overlies the course of the dorsal sensory branch of the ulnar nerve so it can be identified and protected. The extensor retinaculum is incised over the ECU tendon from proximal to distal. No cases of late instability have been reported after surgical release of the ECU tendon from its sheath.²⁰

In cases of acute instability, on the other hand, a 6-week period of immobilization in a pronated long-arm splint is tried first. In cases refractory to initial splinting or for recurrent episodes of symptomatic dislocation, surgical reconstruction of the ECU tendon sheath with the extensor retinaculum is preferred (Fig. 71.6). The tendon is immobilized in a long-arm splint in pronation for 6 weeks after surgery, followed by a period of motion recovery and progressive strengthening exercises.

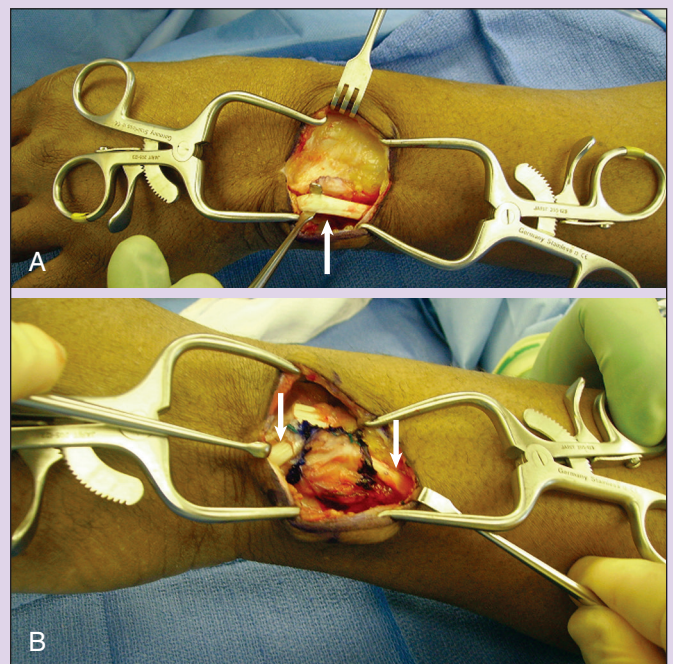


Fig. 71.6 (A) An unstable extensor carpi ulnaris (ECU) tendon is visible outside the normal tendon sheath (arrow). (B) ECU tendon sheath reconstruction using available extensor retinaculum. Notice that the tendon is placed radial to the ulnar head (arrows).



Fig. 71.7 A standard commercially available wrist brace that is useful for many types of wrist tendinopathies. Such a wrist brace is a mainstay of nonoperative care.



Fig. 71.8 An injection of a corticosteroid as adjunctive treatment for flexor carpi ulnaris tendinosis. Care must be taken to avoid inadvertently injecting the ulnar neurovascular bundle.

ECU tendinitis should also be considered. MRI may be helpful but its findings must be correlated with history and physical examination because incidental abnormalities are commonly reported.³⁸ A diagnosis of FCU tendinopathy is probable with a history of repetitive or forceful combined wrist flexion and ulnar deviation, along with clinical examination findings of tenderness over the FCU tendon, exacerbation of pain against resistance, and irritation with passive wrist extension. A trial of NSAIDs and use of a splint or brace, coupled with activity modification, are warranted (Fig. 71.7). Corticosteroid injection may be attempted for more severe cases (Fig. 71.8). If simple conservative measures are not helpful, then adjunctive studies, including MRI, may be warranted to help differentiate FCU tendinopathy from other sources of ulnar-sided wrist pain.¹⁰ This entity rarely requires surgical intervention for alleviation of symptoms.

RESULTS

Treatment results for tendinopathy are usually quite good; full resolution of symptoms is expected by approximately 3 months



Fig. 71.9 Custom-molded splints may expedite return to play during tendon healing. These splints are made specifically to protect involved tendons while allowing participation in sports.

with nonoperative treatment. Certain athletes may be allowed to play with protective splints or casts during the healing process, with the understanding that doing so poses the risk of further injury or even tendon rupture (Fig. 71.9). If surgery is required, both the surgeon and the athlete must recognize that adequate healing time, as well as postoperative rehabilitation, are important and necessary factors for a safe return to play. Reinjury can occur, especially if return to sports is premature, which then requires “restarting the clock” on healing time.

COMPLICATIONS

Common complications faced during the management of wrist tendinopathies include persistent pain despite treatment and prolonged healing times. Many surgeons consider these conditions very stubborn, and they often fail to respond to typical treatment methods because athletes often have difficulty avoiding activities that aggravate the injury. The complication that causes the most concern, however, is the possibility of a tendon rupture. Overuse of steroid injections has been associated with potential tendon rupture, and the athlete should be counseled about the possibility of rupture whenever injections are used.³⁹ Bilateral extensor pollicis longus tendon ruptures have been described in a hockey goalie after steroid injection.⁴⁰ Steroid injections themselves have their own possible adverse effects, which include subcutaneous fat atrophy and skin hypopigmentation in the vicinity of the steroid administration.

Failure of conservative treatment may simply manifest as persistent pain or may involve tendon thickening or tenosynovitis, which may require surgical débridement for optimal resolution of symptoms. Rupture of affected tendons has been described after tenolysis procedures as well and should be part of the informed consent prior to any tendon surgery.⁴¹

FUTURE CONSIDERATIONS

The future of tendon treatment, especially in the athletic population, is already unfolding, with platelet-rich plasma (PRP) injections playing a controversial but definite role in the attempt to speed recovery of injured tendons. The literature remains mixed, with some studies showing a benefit and other studies showing no difference in tendon healing times with PRP versus placebo.^{42–44} Athletes may actively seek PRP injection, because it has become

a popular sports medicine treatment in recent years, but this treatment has yet to be proven superior to other methods. Injection of other growth factors has been considered to speed healing, but to date, well-designed controlled studies are needed before any definitive recommendations can be made.⁴⁵

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Wolfe SW. Tendinopathy. In: Wolfe SW, Hotchkiss RN, Pederson WC, et al, eds. *Green's Operative Hand Surgery*. 2nd vol. 6th ed. Philadelphia: Churchill Livingstone; 2010:2067–2088.

Level of Evidence:

V

Summary:

This expert author's chapter on tendinopathy in a comprehensive hand surgery textbook covers anatomy, pathophysiology, and the author's preferred treatment options. The chapter includes reviews of multiple level I studies currently available in the literature.

Citation:

VanHeest AE, Luger NM, House JH, et al. Extensor retinaculum impingement in the athlete: a new diagnosis. *Am J Sports Med*. 2007;35:2126–2130.

Level of Evidence:

V

Summary:

In this article, expert authors discuss a newly diagnosed form of wrist impingement involving the extensor retinaculum with wrist extension. Although not a true tendinopathy, impingement of the extensor retinaculum is a valid consideration in the athlete with dorsal wrist pain.

Citation:

Banks KP, Ly JQ, Beall DP, et al. Overuse injuries of the upper extremity in the competitive athlete: magnetic resonance imaging findings associated with repetitive trauma. *Curr Probl Diagn Radiol*. 2005;34:127–142.

Level of Evidence:

V

Summary:

This article includes an anatomic discussion of findings on magnetic resonance imaging in persons with repetitive trauma injuries, including tendinopathy.

Citation:

Allende C, Le Viet D. Extensor carpi ulnaris problems at the wrist—classification, surgical treatment and results. *J Hand Surg [Br]*. 2005;30B:265–272.

Level of Evidence:

V

Summary:

This article provides expert author opinions, classifications, and surgical treatment for extensor carpi ulnaris tendonitis. It provides an excellent overview of extensor carpi ulnaris tendinitis, including subluxation and dislocations.

Citation:

Rossi C, Cellocco P, Margaritondo E, et al. De Quervain disease in volleyball players. *Am J Sports Med*. 2005;33:424–427.

Level of Evidence:

III, retrospective cohort study

Summary:

This article describes the occurrence of de Quervain tendinitis in volleyball players, with specific reference to hand and wrist positioning in the sport leading to tendon inflammation of the abductor pollicis longus and extensor pollicis brevis.

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